

## Hypothesis Testing in Multiple Linear Regression

Once we have estimated the parameters in the model, we face two immediate questions:

1. What is the overall adequacy of the model
2. Which specific regressors seem important

### Test for significance of regression

The test for significance of regression is a test to determine if there is a linear relationship between the response  $y$  and any of the regressor variables  $x_1, x_2, \dots, x_k$

The appropriate hypothesis are:

$$H_0 : \beta_1 = \beta_2 = \dots = \beta_k = 0$$

$$H_1 : \beta_j \neq 0 \text{ for at least one } j$$

Rejection of the null hypothesis implies that at least one of the regressors  $x_1, x_2, \dots, x_k$  contributes significantly to the model.

The test procedure is a generalization of the analysis of variance used in simple regression.

The total sum of squares  $SS_T$  is partitioned into a sum of squares due to regression.

$SS_R$  and a residual sum of squares  $SS_{Res}$ . Thus

$$SS_T = SS_R + SS_{Res}$$

The Analysis of Variance Table is given by:

| Source of Variation | Sum of Squares | df          | Mean Square | F                       |
|---------------------|----------------|-------------|-------------|-------------------------|
| Regression          | $SS_R$         | $k$         | $MS_R$      | $\frac{MS_R}{MS_{Res}}$ |
| Residual            | $SS_{Res}$     | $n - k - 1$ | $MS_{Res}$  |                         |
| Total               | $SS_T$         | $n - 1$     |             |                         |

To test the hypothesis  $H_0 : \beta_1 = \beta_2 = \dots = \beta_k = 0$  we compute the F statistic and reject  $H_0$  if

$$F > F_{\alpha, k, n-k-1}$$

The corresponding sum of squares is given by:

$$SS_{Res} = y'y - \hat{\beta}x'y$$

The regression sum of squares is given by:

$$SS_R = \hat{\beta}x'y - \frac{\left(\sum_{i=1}^n y_i\right)^2}{n}$$

and the total sum of squares is given by:

$$SS_T = y'y - \frac{\left(\sum_{i=1}^n y_i\right)^2}{n}$$

## $R^2$ and Adjusted $R^2$

The  $R^2$  is given by:

$$R^2 = \frac{SSR}{SST}$$

The adjusted  $R^2$  which penalizes us for adding terms that are not helpful is defined as:

$$R_{Adj}^2 = 1 - \frac{SS_{Res} \times (n - 1)}{SST \times (n - p)}$$

## Tests on individual regression coefficients

The hypothesis for testing the significance of any individual regression coefficient such as  $\beta_j$  are:

$$H_0 : \beta_j = 0 \text{ vs } H_a : \beta_j \neq 0$$

If  $H_0 : \beta_j = 0$  is not rejected then this indicates that the regression  $x_j$  can be deleted from the model. The test statistic is for this hypothesis is:

$$t_0 = \frac{\hat{\beta}_j}{s.e(\beta_j)}$$

The null hypothesis is rejected if  $|t_0| > t_{\frac{\alpha}{2}, n-k-1}$

## Confidence Intervals on the Regression Coefficients

To construct confidence intervals estimates for the regression coefficients  $\beta_j$  we will continue to assume that the errors  $\varepsilon_i$  are normally and independently distributed with mean and variance  $\sigma^2$

For  $\hat{\beta}$  it is normally distributed with mean vector  $\beta$  and covariance matrix  $\sigma^2(X'X)^{-1}$

We define a  $100(1 - \alpha)\%$  confidence interval for the regression coefficient  $\beta_j, j = 0, 1, \dots, k$  as

$$\hat{\beta}_j \pm t_{\frac{\alpha}{2}, n-p} \sqrt{s.e(\hat{\beta}_j)}$$

## Prediction of New Observations

The regression model can be used to predict future observations on  $y$  corresponding to particular values of the regressor variables for example  $x_{01}, x_{02}, \dots, x_{0k}$ .

If  $X'_0 = [1, x_{01}, \dots, x_{0k}]$  then a point estimate of the future observation  $y_0$  at the point  $x_{01}, x_{02}, \dots, x_{0k}$  is

$$\hat{y}_0 = x'_0 \hat{\beta}$$

A  $100(1 - \alpha)\%$  prediction interval for this future observation.

$$\hat{y}_0 \pm t_{\frac{\alpha}{2}, n-p} \sqrt{\hat{\sigma}^2 (1 + x'_0 (X'X)^{-1} x_0)}$$

## Assumptions of Multiple Linear Regression

The following are the assumptions for multiple linear regression:

1. Normality
2. Linearity
3. Homogeneity of variance (homoscedasticity)
4. No multicollinearity
5. No autocorrelation

### Normality assumption

Standard linear regression relies on the assumption that the residuals are normally distributed.

The hypothesis tested is given by:

$H_0$  : The residuals are normally distributed.

$H_a$  : Residuals are not normally distributed

We use the `shapiro.test()` function in R to investigate the normality assumption.

In addition, we can use the QQ-plot to check for normality of residuals and also the histogram of the residuals themselves.

We investigate the multiple linear regression for the data set described below

```
# fitting multiple linear regression

model <- lm(mpg ~ disp + hp + wt + qsec, data = mtcars)

knitr::kable(broom::tidy(model))
```

The QQ plot is given in the following figure:

```
library(olsrr)

model <- lm(mpg ~ disp + hp + wt + qsec, data = mtcars)
```

```
ols_plot_resid_qq(model)
```

The QQ-plot, all the residuals should follow the diagonal line if there is no violation of the assumption.

The formal test is to use either the shapiro-wilk test, anderson-darling test

```
model <- lm(mpg ~ disp + hp + wt + qsec, data = mtcars)
ols_test_normality(model)
```

The p-values are greater than 0.05 hence we fail to reject the null hypothesis and conclude that the residuals are normally distributed.

A more informative test is to use the performance package

```
library(performance)
model <- lm(mpg ~ disp + hp + wt + qsec, data = mtcars)
check_normality(model)
```

## Linearity Assumption

We use the residuals vs fitted plot to investigate the assumption.

The ideal situation is to have a horizontal line around zero if there are no violations.

```
library(olsrr)
model <- lm(mpg ~ disp + hp + wt + qsec, data = mtcars)
ols_plot_resid_fit(model)
```

The residuals spread randomly around the 0 line indicating that the relationship is linear.

## Homogeneity of Variance

The variance of the residuals is assumed to be constant. The hypothesis tested is given as:

$H_0$  : Constant variances (Homogeneity)

$H_a$  : Non-constant variance (Heterogeneity)

The formal test is the **Breusch-Pagan** test and the function in R is bptest from the lm package.

```
library(lmtest)
model <- lm(mpg ~ disp + hp + wt + qsec, data = mtcars)
bptest(model)
```

The p-value is 0.7285 which is bigger than 0.05 indicating that we fail to reject  $H_0$

## Multicollinearity

Collinearity happens when two or more explanatory variables are correlated with each other.

However, there is an extreme situation, called multicollinearity, where collinearity exists between three or more variables even if no pair of variables has a particularly high correlation.

The Variance Inflation Factor (VIF) can be used to find how much the variance of a regression coefficient is inflated due to multicollinearity in the model.

The smallest possible value is one, indicating no multicollinearity. A value which exceeds 10 indicates a problematic amount of multicollinearity in the data.

In R we use the `vif()` function from the `car` package.

```
library(car)
model <- lm(mpg ~ disp + hp + wt + qsec, data = mtcars)
vif(model)
```

All the variance inflation values are above 10, suggesting our model doesn't have multicollinearity.

If an explanatory variable has a high VIF, we usually remove that explanatory variable from our model.

## Autocorrelation

The Durbin Watson examines whether the errors are autocorrelated with themselves.

The null hypothesis is:

$H_0$  : No autocorrelation

$H_a$  : There is autocorrelation

We use the `durbinWatsonTest()` function in R to check for autocorrelation.

```
library(car)
model <- lm(mpg ~ disp + hp + wt + qsec, data = mtcars)
durbinWatsonTest(model)
```

The p-value is less than 0.05 indicating that we have autocorrelation present in the data.